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MeMoSA dataset: A multi-country collection of over 30,000 oral mucosa images with clinically labelled lesions

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The rising incidence of oral cancer and associated poor prognosis, primarily due to delayed diagnosis, highlight the urgent need for artificial intelligence tools in clinical detection. However, efforts in this regard are hampered by the lack of large and ethnically heterogenous image datasets of oral lesions with clinically validated diagnoses. To address this gap, oral mucosa images captured with mobile device cameras were collected from cohorts spanning five countries. The images were systematically annotated with lesion type classifications as well as specific clinical diagnoses, then assessed for quality. The diagnoses were verified retrospectively by biopsy, where applicable, or by consensus verification by dental experts. The final dataset consists of 30,039 oral mucosa images supplemented by clinical metadata, made available on the MeMoSA Workbench platform. We believe that the MeMoSA dataset will serve as a significant resource to drive the training, evaluation, and refinement of AI-driven diagnostic algorithms, potentially improving diagnostic accuracy and enabling rigorous benchmarking against clinical expert assessments, for the early detection of oral cancer.

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Background & Summary

In 2022, cancer of the lip and oral cavity (referred to here as 'oral cancer') accounted for 398,485 new cases and 188,230 deaths globally[1]. Asia reported around two-thirds of the global incidence and about three-quarters of the global mortality for the disease. With population growth, the oral cancer burden is expected to increase in the next two decades[2].

Early detection and diagnosis of oral cancer is crucial for improved rates of survival for patients[3–6]. Oral cancer may be preceded by a variety of visible lesions collectively known as oral potentially malignant disorders (OPMDs)[7,8]. OPMDs are routinely detected by clinical oral examination (COE)[9], where early identification and referral can reduce the risk of malignant transformation[10]. A trial conducted in Kerala, India demonstrated that screening by COE could reduce mortality in users of tobacco or alcohol[11], which are established risk factors for oral cancer[12,13]. However, limited access to screening facilities contributes to delayed diagnosis[14]. This problem is exacerbated by the need for a specialist's confirmation of the diagnosis, particularly felt in low-and middle-income countries (LMICs), where the number of or access to specialists is limited[9,15].

Recent research has demonstrated the potential of artificial intelligence (AI) to enable the diagnosis, classification and prediction of the occurrence of oral cancer[16–21]. Where specialty care is limited, AI-powered solutions can provide diagnostic assistance to dentists in primary care or to trained workers and volunteers in community settings. Among such approaches are studies that have examined the use of automated detection on mobile phone-captured images of an oral mucosa[22,23]. However, the diagnostic accuracy that can be achieved with automated oral cancer detection techniques is largely dependent on the availability of public oral image datasets, the lack of which is a key challenge impeding advancement in the field[17,24].

Few oral mucosa image datasets have been made publicly available as suitable sources for the training of automated techniques for oral cancer diagnosis[24]. The first was made available on Kaggle.com in 2020 and included 87 'oral cancer' images of the lip and tongue and 44 'non-cancer' images[25]. However, the methods of image acquisition and metadata were not provided. Two studies have reported the use of 'oral cancer image datasets' that would be available by request. The first study described the use of a dataset with 2,350 mobile phone-captured cheek mucosa images from three clinics in India, where images were labelled as 'Normal' or 'Suspicious' by oral oncology specialists[26]. The second used a large dataset of 44,409 clinical oral images consisting of pathologically validated oral cavity squamous cell carcinoma and normal oral mucosa collected from 11 hospitals in China[27]. Another dataset of 3,000 annotated oral mucosa images collected from a hospital in Sri Lanka, categorised as 'Healthy', 'Benign' or 'OPMD', along with the accompanying patient metadata, was made available under restricted access in 2024[28]. Finally, also in 2024, an annotated dataset of 9,201 intraoral images, differentiating between 'normal,' 'low risk of malignant transformation', and 'high risk of malignant transformation' lesions, was made available for non-commercial use[29].

Existing oral mucosa image datasets are currently limited by small size, ethnic homogeneity, a lack of diversity in geographical origin, and constrained availability. Further, existing datasets differentiate mainly between non-cancerous, suspicious for cancer, and benign lesions, without specific diagnosis classifications. To address these limitations and to facilitate efforts in the development of effective algorithms for the detection of oral cancer, we introduce the MeMoSA (Mobile Mouth Screening Anywhere) dataset of oral mucosa images, with expert-verified lesion classifications and referral recommendations for each image.

In 2023, we described the methods for annotation of oral lesion images for the initial development of the MeMoSA image repository, then hosting 2,474 images[30]. Now, the dataset includes 30,039 images of oral lesions or normal oral mucosa collected across 5 countries, along with the accompanying metadata at patient- and lesion-level.

Subsets of these images have been used in previous research for the development of automated algorithms for early detection of oral cancer, as part of the MeMoSA project. In 2020, Welikala *et al.* reported the assessment of deep learning-based approaches for the automated identification of oral lesion-containing images and images of lesions requiring clinical referral[31]. Expanding on this work, a published conference paper described the binary classification of 'referrable' or 'non-referrable' oral mucosa images with deep learning[32]. Further, a mouth landmark detection model was developed to guide the classification of oral lesions into referral decision classes[33]. Also reported was the use of clinically guided trainable soft attention to improve decision making in the automated identification of oral lesions[34].

We believe this diverse dataset of oral cavity images, stored on the MeMoSA database and hosted on the MeMoSA Workbench platform, will be of use to engage researchers of various disciplines in understanding oral cancer and in furthering the development of automated oral cancer detection tools. In this article, we describe the methods for the collection, processing and annotation of the MeMoSA dataset (Fig. 1), and the specifics of the dataset in detail.

Methods

The 30,039 oral mucosa images of the MeMoSA dataset were collected from 6,426 individuals aged 18 and above, from 2015 to 2024, from cohorts in Malaysia, India, Indonesia, Sri Lanka, and Nepal. Participants were recruited through oral medicine or oral surgery clinics, including in-person and through teleconsultation, and from community screening programmes. Recruitment was conducted by clinicians, study coordinators, or healthcare volunteers who invited eligible adults to participate in the study and obtained informed consent for participation and data sharing at the time of collection. Written informed consent was obtained in-person. For data collected through teleconsultation, consent was recorded through a digital consent form. Where participants were unable to provide reasonable consent themselves, parents or guardians of the individual could provide informed consent on their behalf, with a witness present. The images collected were taken by dental experts: dentists or dental specialists, which include those trained in oral medicine or pathology and oral and maxillofacial surgeons at tertiary referral centres[35,36], by healthcare volunteers as part of a community screening program in villages[37], by study coordinators, or by patients themselves as part of a teleconsultation service[38].

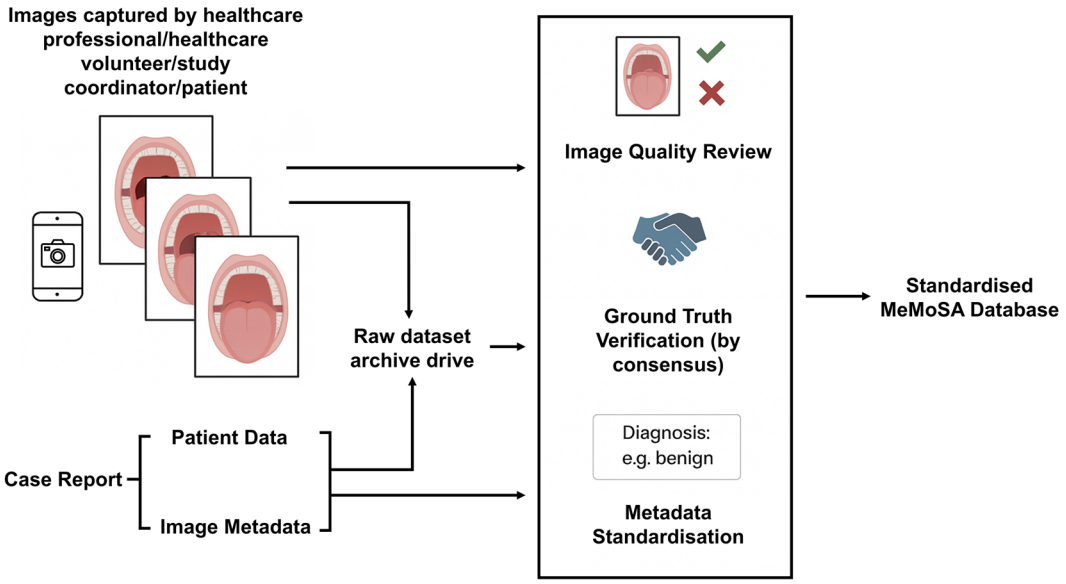


Fig. 1 Schematic diagram showing steps of dataset workup. Images and the corresponding patient and image data collected were reviewed, verified, and standardised by dental experts. Data collected without the use of the MeMoSA app were temporarily uploaded to a raw archive before processing. The standardised images and associated metadata are stored in the MeMoSA database.

One or more images of each individual's oral mucosa were collected, with the intent of capturing various regions of the oral mucosa from specific angles. These regions include the right and left buccal mucosa, the upper and lower labial/sulcus and gingiva, the floor of the mouth, the dorsum and lateral tongue, and the palate. The images were captured using a mobile phone camera with a minimum resolution of 13-megapixels. This requirement was selected based on findings from previous research that revealed that 13-megapixel resolution was sufficient for obtaining high concordance in referral decisions made from oral mucosa images when compared to referral decisions made with in-person COE[36]. Images were captured with the mobile device's camera flash enabled.

The associated metadata collected from each individual included, along with the clinical diagnosis and referral decision provided, information on the individual's demographics, relevant routine clinical information, their reported risk habits[12,13], and, where available, the results of diagnostic biopsy tests. In certain cases, biopsy results are unavailable as the data was captured and recorded before diagnostic biopsy tests were carried out within the clinical pathway. In other cases, the results of the biopsy tests, though performed, were not provided by the cohort contributing the data and are, therefore, not available with the MeMoSA dataset.

The metadata was extracted from its recorded format and stored, initially, on an Excel sheet for manual processing and standardisation before being uploaded for storage on the MeMoSA database. Standardization included structuring of the data with consistent formatting, spelling, and terminology, and verification of the diagnostic labels, described in 'Technical Validation' below. Following upload, the images of the MeMoSA dataset were stored in PostgreSQL, with both the backend and frontend services hosted on an AWS EC2 instance running CentOS, while the domain-name system (DNS) and routing were managed by Cloudflare.

Data collection. *Tertiary referral centres.* 12,264 oral mucosa images were collected from 4,339 individuals referred to or attending long-term follow-up appointments at oral surgery or oral medicine clinics in eight different sites. In Malaysia, these were Universiti Malaya and Ministry of Health referral centres. Other sites include Universitas Trisakti Indonesia, JSS Academy of Higher Education and Research India, Saveetha University India, BP Koirala Institute of Health Sciences Nepal, and the University of Peradeniya Sri Lanka. In previously reported feasibility studies, Haron *et al.* described the methods for the collection and use of a subset of images from the MeMoSA dataset from the Malaysian sites[35,36]. Rajendran *et al.* reported in brief, also, collection methods for a further subset of images in a study intended to evaluate disease monitoring through teleconsultation[38].

A set of images of the regions of the oral mucosa was captured by the study coordinators or dental experts, during or after in-person consultations, or by the patient themselves for teleconsultation services, with a mobile device that met the minimum resolution requirements. The images were collected either through the MeMoSA App[39] or using the camera application on the mobile device. Images captured without the use of the MeMoSA App were manually uploaded for storage in a Google Drive, from which they were subsequently extracted for storage on the MeMoSA database, while images captured using the app were automatically uploaded to the MeMoSA server and stored on the MeMoSA database.

The patients' demographic, risk habit, and relevant routine clinical data were recorded. A clinical diagnosis was provided by a dental expert. These diagnoses were provided either outside routine care, based on COE findings, or remotely as part of clinical service via teleconsultation, to determine if an oral lesion was present, the

type of oral lesion present, if any, and if the lesion was suspicious for OPMD or oral cancer. A referral decision, if required, was determined. When lesions were biopsied for diagnostic reasons, biopsy results, if available, were recorded also. This data was recorded on an electronic data collection form or on the MeMoSA app.

In cases where, following consultation, the patient was required to attend a follow-up appointment at a clinic, the dental specialist at the appointment could, following COE or treatment, capture a further set of images to be added to the dataset, using a device that met the resolution requirements.

Community screening. 17,775 images in the MeMoSA dataset were collected during community screening carried out in villages in Malaysia. Images were collected from 224 individuals from a village in Peninsular Malaysia in conjunction with routine screening programs conducted by the Ministry of Health, Malaysia. The methods of collection of this data were reported by Haron *et al*. in a mobile health feasibility study that used a subset of the data captured[37]. Images were also collected from a further 1,863 individuals, from villages in Sarawak, Malaysia. The MeMoSA App was used to capture the oral mucosa images and associated patient data[39].

In the Sarawak sites, the MeMoSA App was used primarily by healthcare volunteers trained to collect oral mucosa images and patient data, while in the screening program carried out in Peninsular Malaysia, images and patient data were captured by dentists on-site. Within the app, the healthcare volunteers or dentists were directed to take a set of oral mucosa images, capturing all regions of the mouth using a mobile phone with a 13-megapixel camera. These were uploaded automatically to the MeMoSA server for storage on the MeMoSA database.

The healthcare volunteers and dentists recorded the demographic, risk habit, and clinically relevant information for the patients being screened using the MeMoSA app. The patients' images and information were then remotely reviewed by a dental expert, to provide a clinical diagnosis and a referral decision, if required as part of clinical service delivery. At follow-up appointments, additional images could be captured by dental specialists. When lesions were biopsied for diagnostic purposes, biopsy results were recorded and added to the patient's data.

Image quality assessment. The quality of the images was manually assessed by a single dental expert and labelled as either "good" or "poor" quality images. If images were labelled "poor" quality a reason for "poor" quality, where available, was noted with the image metadata, according to the criteria outlined below. The images that were labelled "poor" quality were not removed from the MeMoSA dataset, as, for most, a clinical diagnosis for the area captured in the oral mucosa images could still be determined, demonstrating their value as part of a diagnostic benchmark dataset. This includes potential use in training algorithms to recognise poor quality images. Such images may, further, be representative of those that might be provided by end-users, or patients, in a real-world scenario via mobile device capture. Two images for which the quality was too poor to make a conclusive diagnosis were also included, as they may contribute value as image quality benchmarks.

The following criteria were used for determining a "poor quality image:

1. Images where the area of interest within the oral mucosa was not in frame
2. Images with visual artefact(s) present
3. Images that captured extraoral regions
4. Images that were out of focus
5. Images that were dark
6. Overexposed images

Images with extraoral regions that revealed identifiable facial features were manually cropped so that only oral regions remained visible to reduce the risk of participant identification.

Ethics approval. Ethics approval for data collection was obtained from the following review boards and ethics committees:

- Medical Ethics Committee, Faculty of Dentistry, Universiti Malaya, Malaysia (Reference Nos: DF OS1902/0011(L), DF OS1902/0011(L)/1970, DF OS2210/0022(L), DF OS2210/0022/24109(L), DF OS2016/0069/20132(L), DF OS2016/0069/20132/20154(L))
- Medical Research and Ethics Committee, Ministry of Health Malaysia, Malaysia (Reference Nos: NMRR-15-763-25620 (IIR), NMRR ID-23-01575-WCR (IIR))
- Ethical Review Board, Nepal Health Research Council (NHRC), Nepal (Reference No: 101/2019)
- Ethical Review Committee, Faculty of Dental Sciences, University of Peradeniya, Sri Lanka (Reference No: ERC/FDS/UOP/I/2019/04)
- Ethics Committee for Health Research, Faculty of Dentistry, Trisakti University, Indonesia (Reference Nos: 249/S3/KEPK/FKG/3/2019, 010/S3/KEPK/FKG/08/2021, 060/S3/KEPK/FKG/1/2024)
- JSS Dental College & Hospital Institutional Ethics Committee, JSS Academy of Higher Education & Research, India (Reference No: 06/2019)
- Scientific Review Board, Saveetha Dental College & Hospitals, Saveetha Institute of Medical and Technical Sciences, India (Reference No: SRB/SDC/Faculty/22/OMED/038)
- Institutional Human Ethical Committee, Saveetha Dental College & Hospitals, Saveetha Institute of Medical and Technical Sciences, India (Reference No: IHEC/SDC/FACULTY/22/OMED/411)

All participants provided informed consent for participation in the broader MeMoSA project, image capture, and data sharing for research, teaching, and publication purposes. Where applicable, agreement was obtained from the original data providers for the sharing of data. Subsets of the data are made available subject to differing

licensing and data-sharing conditions in accordance with the terms of informed participant consent and institutional data-sharing agreements.

Data Records

All the data records of the MeMoSA dataset are deposited on the MeMoSA Workbench platform, listed as separate country- or organisation-level collections: Malaysia[40]; Saveetha, India[41]; JSS, India[42]; Sri Lanka[43]; Indonesia[44]; and Nepal[45]. The images can be accessed on the MeMoSA Workbench and are available for download in JPG, JPEG or PNG format.

Throughout the dataset, images are assigned IDs using the format '*Country_Cohort_PatientID_ImageNumberbyPatient_ImageNumberbyCohort.FileFormat*'. Each country from which images were collected was assigned a 3-letter code (Malaysia: *MYS*, India: *IND*: Sri Lanka: *LKA*, Indonesia: *IDN*, Nepal: *NPL*). Similarly, each cohort was assigned a unique 3-letter code. Each individual or patient was given an anonymised and unique patient ID, and images were numbered by patient and by each cohort, overall.

The metadata, at patient- and lesion-level associated with the image is included as a comma-separated values (CSV) file. The rows of the CSV file represent consecutive images with the following attributes:

- Image ID
- Age (of patient, at time of image capture)
- Gender
- Ethnicity
- Smoking status
- Chewing betel quid status
- Alcohol consumption status
- Duration of smoking habit
- Duration of betel quid chewing habit
- Duration of alcohol consumption habit
- Lesion type
- Clinical diagnosis
- Referral decision
- Image captured by (patient, clinician, healthcare volunteer, or study coordinator)
- Clinical presentation of the lesion
- Chief complaint
- History of presenting complaint
- Biopsy results, if any
- Other investigations or tests results, if any
- Medication history
- Medical history
- Use of sodium lauryl sulphate (SLS) containing toothpaste
- The SLS containing toothpaste used, if any
- Use of other oral hygiene products
- Type of other oral hygiene product used, if any
- Comments from the image provider
- Image quality
- Reasons for poor quality, where available

Data Overview

Table 1 provides a summary of the main characteristics of the dataset, outlining the number and proportion of images included by lesion type. Lesion type categories exhibit similar distribution across gender, while images from individuals above 40 years of age make up a larger proportion of all lesion type categories. Graphical representations of the data distribution by gender and age groups across lesion type categories are included in the Supplementary Information (Supplementary Fig. 1).

In addition, descriptive statistics of the image characteristics, including pixel dimensions, aspect ratio, resolution and file size, along with algorithmic evaluation of image quality using the Blind/Reference less Image Spatial Quality Evaluator (BRISQUE) no-reference metric[46] using the *brisque* Python package[47], are reported in Table 2 for a comprehensive overview of the technical properties of the dataset.

Technical Validation

Unifying clinical diagnoses. All cases were clinically diagnosed by dental experts. 165 unique clinical diagnoses were provided for the oral mucosa images collected across the MeMoSA dataset. High variability was observed within these original diagnoses, including misspellings, different oral medicine terminologies, and multiple lesion diagnoses for a single image.

To standardise the diagnostic metadata associated with the dataset, we unified the original clinical diagnoses to 122 "overall" diagnoses[48], which inform the referral recommendations provided for each image (Supplementary Table 1). These diagnoses were further unified to form six lesion type classes. The six lesion type classes differentiate between diagnoses of oral cancer, OPMD, benign lesions, developmental anomalies, normal anatomical variants, and normal oral mucosae with no lesion present. More detailed descriptions of these lesion type classes are provided with the usage notes below.

Lesion Type	Cancer (n = 619) n (%)	OPMD (n = 5076) n (%)	Benign (n = 3054) n (%)	DA (n = 314) n (%)	NAV (n = 721) n (%)	No Lesion (n = 20253) n (%)	N/A* (n = 2) n (%)	Total (n = 30039) n (%)
Gender								
Female	182 (29.4)	2054 (40.5)	1344 (44.0)	124 (39.5)	316 (43.8)	12022 (59.4)	1 (50.0)	16043 (53.4)
Male	375 (60.6)	2795 (55.0)	1521 (49.8)	186 (59.2)	385 (53.4)	8198 (40.4)	1 (50.0)	13461 (44.8)
N/A**	62 (10.0)	227 (4.5)	189 (6.2)	4 (1.3)	20 (2.8)	33 (0.2)	0 (0.0)	535 (1.8)
Age group								
<40	30 (4.8)	1068 (21.0)	769 (25.2)	123 (39.2)	240 (33.3)	6539 (32.3)	0 (0.0)	8769 (29.2)
≥40	518 (83.7)	3759 (74.1)	2067 (67.7)	186 (59.2)	460 (63.8)	13680 (67.5)	2 (100.0)	20672 (68.8)
N/A**	71 (11.5)	249 (4.9)	218 (7.1)	5 (1.6)	21 (2.9)	34 (0.2)	0 (0.0)	598 (2.0)
Country								
Malaysia	46 (7.4)	1399 (27.6)	1138 (37.3)	210 (66.9)	458 (63.5)	19882 (98.2)	2 (100.0)	23135 (77.0)
India	99 (16.0)	848 (16.7)	200 (6.5)	66 (21.0)	92 (12.8)	0 (0.0)	0 (0.0)	1305 (4.3)
Indonesia	64 (10.4)	67 (1.3)	207 (6.8)	23 (7.3)	16 (2.2)	1 (0.0)	0 (0.0)	378 (1.3)
Nepal	31 (5.0)	93 (1.8)	120 (3.9)	3 (1.0)	3 (0.4)	1 (0.0)	0 (0.0)	251 (0.8)
Sri Lanka	379 (61.2)	2669 (52.6)	1389 (45.5)	12 (3.8)	152 (21.1)	369 (1.8)	0 (0.0)	4970 (16.6)
Smoking								
No	359 (58.0)	3929 (77.4)	2273 (74.4)	235 (74.8)	514 (71.3)	15503 (76.6)	2 (100.0)	22815 (76.0)
Occasionally	0 (0.0)	22 (0.4)	20 (0.7)	3 (1.0)	8 (1.1)	432 (2.1)	0 (0.0)	485 (1.6)
Yes	179 (28.9)	858 (16.9)	379 (12.4)	51 (16.2)	165 (22.9)	4288 (21.2)	0 (0.0)	5920 (19.7)
N/A**	81 (13.1)	267 (5.3)	382 (12.5)	25 (8.0)	34 (4.7)	30 (0.1)	0 (0.0)	819 (2.7)
Chewing Betel Quid								
No	225 (36.3)	2933 (57.8)	2284 (74.8)	269 (85.7)	630 (87.4)	18079 (89.3)	2 (100.0)	24422 (81.3)
Occasionally	0 (0.0)	29 (0.6)	34 (1.1)	9 (2.9)	5 (0.7)	1028 (5.1)	0 (0.0)	1105 (3.7)
Yes	320 (51.7)	1864 (36.7)	354 (11.6)	11 (3.5)	52 (7.2)	1116 (5.5)	0 (0.0)	3717 (12.4)
N/A**	74 (12.0)	250 (4.9)	382 (12.5)	25 (7.9)	34 (4.7)	30 (0.1)	0 (0.0)	795 (2.6)
Alcohol								
No	415 (67.0)	4255 (83.8)	2273 (74.4)	202 (64.3)	521 (72.3)	13918 (68.7)	2 (100.0)	21586 (71.9)
Occasionally	1 (0.2)	44 (0.9)	175 (5.7)	61 (19.4)	114 (15.8)	5066 (25.0)	0 (0.0)	5461 (18.2)
Yes	122 (19.7)	504 (9.9)	227 (7.5)	26 (8.3)	52 (7.2)	1239 (6.1)	0 (0.0)	2170 (7.2)
N/A**	81 (13.1)	273 (5.4)	379 (12.4)	25 (8.0)	34 (4.7)	30 (0.2)	0 (0.0)	822 (2.7)

Table 1. Summary of patient-level dataset characteristics by lesion type (OPMD, oral potentially malignant disease; DA, developmental anomaly; NAV, normal anatomic variant). *Images without ground truth labels, as image quality was too poor for assessment and verification. **Parameters were not reported by the data contributor.

Parameter	Mean ± SD	Median (IQR)	Minimum	Maximum
Image width (pixels)	1696.4 ± 1204.4	981 (979–2496)	95	8000
Image height (pixels)	1823.8 ± 871.7	1440 (1440–2448)	95	6000
Aspect ratio (width/height)	0.9 ± 0.3	0.7 (0.7–1.0)	0.4	2.7
Image resolution (dpi)	72.6 ± 4.3	72 (72–72)	72	220
File size (KB)	671.2 ± 915.5	170.5 (123.1–1125.4)	2.8	17322.7
BRISQUE score	24.5 ± 14.2	22.7 (13.3–34.8)	−25.8	155.8

Table 2. Descriptive statistics of image characteristics.

For images where there was an abnormality in the oral mucosa but an inability to assign a single diagnosis that satisfied the criteria for a specific oral lesion, an "overall" diagnosis label called "Others" was added, with a recommendation to refer cases categorised as such for further investigation.

The "overall" diagnosis, lesion type, and referral recommendation associated with each image are reported in the metadata of the MeMoSA dataset.

Ground truth verification by consensus. The original clinical diagnoses provided for each oral mucosa image are referred to here as ground truth labels. Where biopsy results were available (323/30,039, 1.1% of total images in dataset), these were used to retrospectively verify the ground truth labels. For all remaining images, ground truth verification was carried out by dental experts, by reviewing the oral mucosa images and the ground truth labels provided (Fig. 2). Following verification of the ground truth labels, the corresponding "overall" diagnosis and lesion type labels were added to the metadata associated with each image. If a referral decision had not

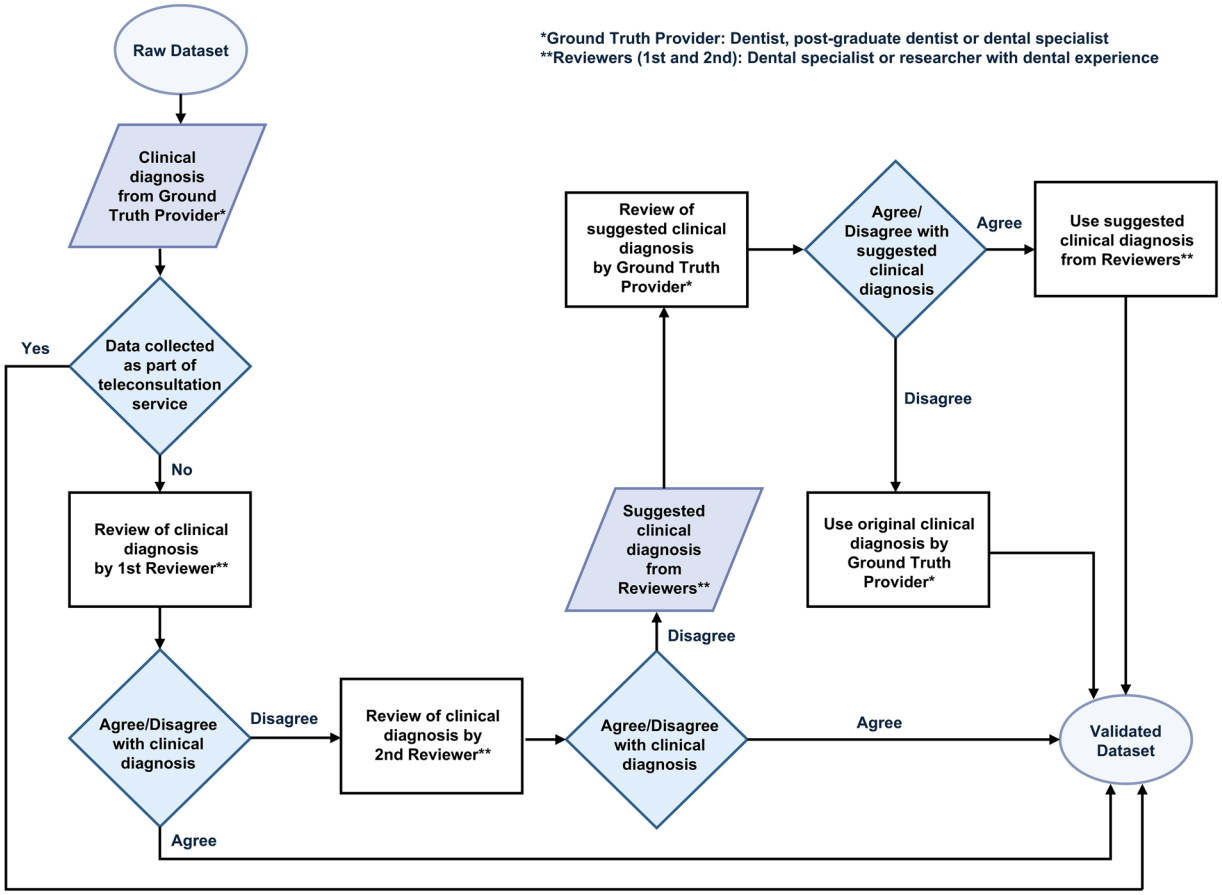


Fig. 2 Schematic diagram showing workflow of verification of ground truth labels. Clinical diagnoses provided for each oral mucosa image in the raw dataset were reviewed by experts in a verification process to generate the diagnosis labels for the validated MeMoSA dataset. Images collected for teleconsultation services were added directly with their clinical diagnoses to the validated dataset and did not undergo this process.

originally been made, the appropriate recommendation was also added to the metadata based on the ground truth label. Where more than one diagnosis had been made for a single image, lesion type and referral decision labels were assigned based on the diagnosis with higher clinical risk.

Ground truth labels were first reviewed by a single dental expert, the first reviewer. If the first reviewer disagreed with the ground truth label, a second review would be carried out by another dental expert, and a consensus reached between the reviewers. If both disagreed with the ground truth label, a suggested clinical diagnosis was provided by consensus agreement. These suggested diagnoses were shared with and reviewed by the ground truth providers, who had supplied the original clinical diagnosis. If the ground truth providers agreed with the suggested diagnosis, the suggested ground truth label replaced the original in the image metadata. If the ground truth providers disagreed with the suggestion, the original ground truth label was retained.

Clinical diagnoses for images that were collected as part of clinical service delivery, namely for remote teleconsultation (17,291/30,039, 57.6% of total images in dataset), were standardised to assign "overall" diagnosis, lesion type, and referral decision labels, but did not undergo the additional multi-reviewer verification process by consensus. In such cases, image acquisition and diagnosis were part of the same clinical encounter, and the diagnosis was provided remotely as part of patient care based on the clinical presentation from multiple images captured during the encounter. Other images, collected outside routine care and subsequently assigned a ground truth label, required independent verification, as expert reviewers relied solely on retrospective validation of the recorded data. Previous work has demonstrated high diagnostic concordance between remote assessments through the MeMoSA App and in-person COE, supporting confidence in the reliability of these diagnoses[36].

Metadata that had undergone ground truth verification was said to have been verified and standardised for addition to the MeMoSA database as part of the MeMoSA dataset.

Usage Notes

This dataset includes images of normal oral mucosa and a wide variety of oral lesion types from populations in South and Southeast Asia, where the oral cancer disease burden and practice of attributable risk factors are high[1], with representative examples of all the oral lesions that are practically relevant in these populations and would be commonly encountered in clinical practice[49].

This dataset is intended to improve the potential of algorithms as preliminary diagnostic tools, which take into account clinically relevant information that would influence decision making in routine practice. Furthermore, the dataset includes specific diagnoses of the oral lesions captured as well as differentiating between cancerous, suspicious, and benign lesions, mimicking clinical practice. The referral decisions provided reflect routine clinical practices according to diagnoses in the countries from which the data was collected, though these recommendations may differ in other regions.

We note that the dataset contains relatively diverse data, given the collection of images from cohorts across 5 countries. However, the generalisation of algorithm-based lesion classification or diagnosis to clinical practice depends on the demographic agreement of the dataset to the clinical population. This could lead to misdiagnosis in cohorts underrepresented in the dataset. We also caution against overfitting algorithms to particular subsets or cohorts of this dataset.

Description of diagnostic categories. The following description of the six oral lesion type classes used in the metadata is meant for users of the MeMoSA dataset who are not familiar with the oral disease literature:

Oral cancer. The term "oral cancer" is commonly used to refer to oral squamous cell carcinomas, malignant neoplasms typically originating from the mucosal epithelium of the oral cavity. A smaller proportion arise in minor salivary glands or from metastasis. Early oral cancer lesions or carcinomas typically present as painless red, white, or speckled patches, often indistinguishable from potentially malignant lesions, while a small proportion may be ulcerated[50]. Advanced lesions may develop into raised nodules, become ulcerated, or both, developing into indurated ulcers with rolled borders, at which stage the carcinoma is usually several months old. Early oral cancer lesions are generally painless, though some may cause burning or stinging pain. Larger carcinomas may lead to neuropathic pain, paraesthesia, or referred pain in the ear. Pain intensity typically increases as the disease progresses.

Oral potentially malignant disorder (OPMD). Oral potentially malignant disorders (OPMD) refer to a heterogenous group of clinically defined conditions that are associated with the risk of development into oral cancer, most of which give rise to clinically visible lesions[48]. These disorders are believed to arise from widespread genetic alterations in the affected tissue, a phenomenon known as field change or cancerization, increasing the risk of cancer developing in multiple sites within the affected region. OPMDs can have a varied clinical presentation, including predominantly red oral mucosal areas that cannot be definitively diagnosed, white plaques of uncertain risk after excluding other definable or clinically recognisable conditions that carry no risk for malignant transformation, and other identifiable mucosal changes, which carry an increased risk for oral cancer.

Benign. Benign oral lesions refer to non-cancerous growths or changes in the oral cavity. These lesions are typically localized, do not invade surrounding structures, and do not metastasize to other parts of the body[51–53]. They can be congenital or acquired, and while they generally do not pose a serious threat, some may require treatment if they cause discomfort or interfere with function.

Developmental anomaly (DA). Developmental anomalies in the oral cavity, are conditions that result from disruptions during the normal developmental processes of the oral and craniofacial structures[54,55]. These anomalies can be congenital or may develop during the growth phases of childhood and adolescence. They can affect any part of the oral cavity, including teeth, soft tissues, or bone. Many developmental anomalies are genetic, but some may result from environmental influences such as infections or trauma during development.

Normal anatomic variant (NAV). Normal anatomic variants refer to naturally occurring features in the oral cavity that are considered part of typical human variation[56–59]. These variants do not represent disease or pathological conditions, but rather reflect diversity in anatomy, which can be influenced by genetic, environmental, and lifestyle factors. These do not require treatment and are typically of no clinical significance unless they cause functional impairment or aesthetic concerns.

No lesion. An oral cavity with normal mucosae and no identifiable lesions.

Accessing the MeMoSA dataset. The MeMoSA dataset is available on the MeMoSA Workbench platform. The data contributed by the BP Koirala Institute of Health Sciences, Nepal, the JSS Academy of Higher Education and Research, India, and the University of Peradeniya, Sri Lanka are licensed under a Creative Commons Attribution 4.0 International (CC BY 4.0) license. The data contributed by Trisakti University, Indonesia, the Saveetha Institute of Medical and Technical Sciences, India, Universiti Malaya and the Ministry of Health, Malaysia are licensed under a Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) license. Due to differing licensing, data from each institution are listed as separate collections. However, upon request, researchers will be granted access to all collections as part of the full dataset.

Before accessing this data, researchers will need to complete the following steps:

1. Register for an account on the MeMoSA database (https://workbench.memosa.my)
2. Ensure the account is approved by MeMoSA Workbench administrators
3. Submit a request for restricted access data, detailing intended data use
4. Agree to the Data Use Agreements (data from certain sources may have additional conditions for use)
5. Receive request approval

Users of the MeMoSA dataset must only use the data for agreed upon purposes described in the Data Use Agreement. Users of the dataset should acknowledge the contributions of the original authors and data contributors when using the data by citing this article and the repository link for the dataset.

As we plan to continue expanding the MeMoSA dataset in the future, additional data releases will be announced and made available on the MeMoSA Workbench platform. Any issues or feedback can be submitted using the contact details listed on the platform, and the administrators will address these promptly.

EXIF-data of the images were manually cleared to remove embedded geographical, temporal, and phone model data. This was done to prevent potential disclosure of sensitive of identifiable information, following established data protection practices. While the authors acknowledge that EXIF information can aid in machine learning training, its removal was necessary to enhance participant privacy and ensure compliance with ethical data-sharing standards. The data in the MeMoSA dataset has been reviewed to minimise the presence of identifiable facial images. Informed consent was obtained from participants for the sharing and use of their data, which has been processed to minimise identifiability, with the retention of potentially identifiable information in the dataset metadata, including individuals' gender, ethnic group, and demographic data.

Data availability

The MeMoSA dataset[40–45] is available via the MeMoSA Workbench platform (https://workbench.memosa.my).

Code availability

No custom code was used.

Received: 28 July 2025; Accepted: 27 February 2026;
Published online: 07 March 2026

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Acknowledgements

We would like to acknowledge the postgraduate students, staff, dental officers and dental specialists from the oral medicine departments of Universiti Malaya, University of Peradeniya, BP Koirala Institute of Health Sciences, Trisakti University, JSS Academy of Higher Education and Research, Saveetha Institute of Medical and Technical Sciences, Hospital Tengku Ampuan Rahimah, Hospital Serdang, and the Ministry of Health Malaysia for their valuable assistance in the collection, curation and sharing of this data. We wish to acknowledge the contributions of time and data from the MeMoSA project participants. We thank the supporters of Cancer Research Malaysia. Cancer Research Malaysia is a not-for-profit research organization. We are committed to an understanding of cancer prevention, diagnosis, and treatment through a fundamental research program.

Author contributions

S.C. wrote the manuscript and contributed to data processing. A.N. contributed to design and data processing. W.J.S., H.Y.L. and S.N.S. contributed to data processing. D.S.R. and S.H.L. contributed to data analysis and interpretation. C.S.C. and W.M.T. contributed to conceptualisation. R.D.J., J.R., R.A. and K.P. contributed to data acquisition, analysis and interpretation. A.M., S.M.I., Z.M.Z., C.C.T., Y.C.G., S.W.C., N.I.Z., M.K.H., K.S., K.K., N.F.Z.A., N.P.H.K. and T.H.M. contributed to data acquisition. C.S.L. contributed to conceptualisation and design. A.R.K., T.G.K. and R.Z., contributed to design and data acquisition. S.R. contributed to design, data acquisition, and data analysis and interpretation. S.C.C. contributed to conceptualisation, design, data acquisition, and data analysis and interpretation. All authors provided editorial feedback during the revision process and participated in preparation of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41597-026-06998-7.

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